

ECON 21030: Honors Econometrics

Statistics Review

Spring Quarter 2024

Sampling, estimators, estimands

In the last lecture, we reviewed how to model and describe uncertain outcomes using random variables. Now we are going to turn to the task of learning about random variables from *data*. To do so, we will use the notion of a [sample](#). A sample is a collection of random variables. For most of the course (but not all), we will consider samples that are [identically and independently distributed \(iid\)](#). This means that a sample of size N consists of a set of random variables $X_1, \dots, X_N$, each of which is independent and distributed according to the same F . A simple example would be a series of heads or tails from flipping a coin, or the recorded height from a random sample from a very large population of people. You can see already that the iid assumption is usually an approximation, but it can be a reasonably good one.

The random variables in the sample may themselves be k -dimensional. We may, for example, collect information on both the height and weight of the individuals we sample from this large population. It is useful to collect these random variables into the N by K matrix $\mathbf{X}$. Each row of this matrix is a k -dimensional random vector, with each row identically and independently sampled from a multivariate distribution F .

A sample is itself random. Realizations of this random vector are what you have in your [dataset](#). Our goal now will be to define functions of the sample, which we call [estimators](#), that tell us something about distributions of the underlying random variables. Since estimators are functions of the sample, they are also random variables. They have their own distributions, moments, etc. In practice any given dataset will give rise to a realization of the estimator, which you can call an [estimate](#).

The goal in estimation is to learn about some feature of the distribution F . We call the target of the estimator the [estimand](#). That is, the object you're hoping the estimator will tell you about—the thing “to be estimated” (leaning on the future passive participle in the latin here).

This is all pretty abstract, so let's make things more concrete. Suppose we have an iid sample $X_1, \dots, X_N$ from F , an unknown distribution with finite first and second moments. One potential estimand we might be interested in is the mean of F , or $E[X_i]$, which we will call μ . The most common estimator for μ is the sample mean:

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N X_i$$

The “hat” ($\hat{x}$) notation is convention that means you are defining an estimator of the estimand under the hat (x). The sample mean is also special enough that it gets own notation convention and is typically written as $\bar{X}_N$ or just $\bar{X}$.

You can see that $\hat{\mu}$ is a function of the sample (i.e., $\hat{\mu} = h(X_1, \dots, X_N)$). The sample is a collection of random variables. This makes $\hat{\mu}$ itself a random variable with its own induced distribution. You should be able to use to properties we covered in the last lecture to show, for example, that:

$$E[\hat{\mu}] = \frac{1}{N} \sum_{i=1}^N E[X_i] = \mu$$

$$Var(\hat{\mu}) = \frac{1}{N^2} \sum_{i=1}^N Var(X_i) = \frac{\sigma^2}{N}$$

where σ^2 is the variance of F . The mean of $\hat{\mu}$ is therefore the same as the mean of F , which is why it is a useful estimator for μ . The variability of $\hat{\mu}$ is related to the variance of F and the size of the sample N .

Some people will call an estimand a [parameter](#). For example, they might say that the parameter estimated in the example I just gave is μ . More generally, in econometrics we consider parameters to be the ingredients of economic models that describe how the world works. This model may be very simple. In fact, the model may simply be that X_i is iid $\sim F$. If F has mean μ , μ is indeed an interesting parameter of the model. We will discuss cases later, however, where the parameter of interest may or may not be [identified](#) from the estimands under scrutiny.

There are three properties of estimators that will be important in this class. The first is the [bias](#) of an estimator $\hat{\theta}$ of θ . We call an estimator $\hat{\theta}$ [unbiased](#) for θ whenever $E[\hat{\theta}] = \theta$. The bias itself is defined as $B(\hat{\theta}; \theta) = E[\hat{\theta}] - \theta$. Notice that bias is only defined in relation to the target of estimation, i.e., the estimand. For example, any estimator $\hat{\theta}$ is an unbiased estimator of $E[\hat{\theta}]$ provided that expectation is well defined. Likewise, an estimator that is unbiased for one quantity need not be unbiased for another.

The second property of estimators that is important is their [consistency](#). Roughly, consistency implies that as we collect more data, we expect $\hat{\theta}$ to approach its target θ . Since the estimator is a random variable, we cannot usually guarantee that it will *always* be within some distance of θ with sufficient N , as in the case of the limiting behavior of a non-random sequence you might study in your analysis classes. Instead, we use a weaker notion of convergence and say that a sequence of random variables X_n [converges in probability](#) to X if for every $\epsilon > 0$:

$$Pr(|X_n - X| > \epsilon) \rightarrow 0$$

as $n \rightarrow \infty$. Here, the limit going to zero in n means that for every ϵ and δ there is some n' such that $n > n'$ implies $Pr(|X_n - X| > \epsilon) < \delta$. Though this definition is written for a sequence of scalar random variables, we also say that sequence of random vectors or matrices $\mathbf{X}_n$ converges in probability to $\mathbf{X}$ when it does so elementwise.

When an estimator $\hat{\theta}$ converges in probability to θ , we say it is consistent for θ . Note that θ can be fixed (i.e., a number) or random variable itself. Convergence in probability is denoted as $\hat{\theta} \xrightarrow{p} \theta$, or $\text{plim } \hat{\theta} = \theta$.

A useful way to characterize consistency is to make use of Chebyshev's inequality, which I introduced last lecture and states that $Pr(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$. Letting $k = \epsilon/\sigma$, we can rewrite this inequality as $Pr(|X - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2}$. Thus for a random sequence X_n , we have:

$$Pr(|X_n - E[X_n]| > \epsilon) \leq \frac{E[(X_n - E[X_n])^2]}{\epsilon^2}$$

$$Pr(|X_n - E[X_n]| > \epsilon) \leq \frac{Var(X_n)}{\epsilon^2}$$

Thus clearly if an estimator is unbiased, so that $E[X_n] = \theta$, and has variance that converges to zero (i.e., $Var(X_n) \rightarrow 0$), it is going to be consistent.

This is actually the basis of the [weak law of large numbers](#), which states that for an iid sample $X_1, \dots, X_n$ with $Var(X_i) < \infty$, then $\hat{\mu} = \frac{1}{N} \sum X_i \xrightarrow{p} E[X_i]$. We can see that this holds from our example above, where we showed that $Var(\hat{\mu}) = \sigma^2/N$, which clearly converges to zero in N . By Chebyshev we therefore know that convergence in probability will be satisfied.¹

Students sometimes are confused by the connection between bias and consistency (or lack thereof). The former does not imply the latter, nor vice-versa. For example, we know that $\hat{\mu}^{dumb} = E[X_1]$ is an unbiased estimator $E[X]$. But the variance of $\hat{\mu}^{dumb}$ is simply σ^2 , which is not decreasing in n . This estimator is therefore unbiased but inconsistent.

Likewise, consider $\hat{\mu}^{biased} = \bar{X} + \frac{1}{N}$. This estimator is clearly biased due to the $1/N$ term, which implies $E[\hat{\mu}^{biased}] = E[\bar{X}] + 1/N > \mu$. It is nevertheless consistent for μ because:

$$\begin{aligned} Pr(|\hat{\mu}^{biased} - \mu| > \epsilon) &= Pr(|\hat{\mu}^{biased} - \mu - 1/N + 1/N| > \epsilon) \\ &\leq Pr(|\hat{\mu}^{biased} - E[\hat{\mu}^{biased}]| + 1/N > \epsilon) \\ &\leq \frac{Var(\hat{\mu}^{biased})}{(\epsilon - 1/N)^2} \end{aligned}$$

Since $Var(\hat{\mu}^{biased})$ is simply σ^2/N and thus will go to zero, convergence in probability will clearly hold.

Two very useful facts about convergence in probability come up frequently in econometrics. The first is [Slutsky's theorem](#). Slutsky says that if $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} c$, where c is a constant, then $X_n + Y_n \xrightarrow{p} X + c$. Likewise, $X_n Y_n \xrightarrow{p} Xc$ and $X_n/Y_n \xrightarrow{p} X/c$ so long as $c \neq 0$. We could have shown consistency in example above using Slutsky as well. When $\hat{\mu}^{biased} = \bar{X} + 1/N$, we know that $\bar{X} \xrightarrow{p} \mu$ and $1/N \rightarrow 0$. Thus by Slutsky $\hat{\mu}^{biased} \xrightarrow{p} \mu$. Some students find it helpful to remember Slutsky through the mnemonic that it “puts results together.”

The second fact is known as the [continuous mapping theorem](#). It says that if $X_n \xrightarrow{p} X$, then $g(X_n) \xrightarrow{p} g(X)$ if g is continuous at all x in the support of X . You can see why continuity is important here: even though X_n is getting “close” to X , $g(X_n)$ may not be close to $g(X)$ if g has big jumps.

Continuous mapping is useful because it also “puts results together.” Specifically, we can consider vector random variables and multivariate functions. For examples, suppose that $\mathbf{X}_n = (Y_n, Z_n)'$

¹There is also a *strong* law of large numbers, which implies a stronger notion of convergence than convergence in probability and will not be relevant for our work in this class.

and $g((x_1, x_2)') = x_1 \cdot x_2$. Then if $\mathbf{X}_n \xrightarrow{p} (Y, Z)'$, we have $g(\mathbf{X}_n) \xrightarrow{p} YZ$. Continuous mapping also implies that if a sequence of invertible random matrices $\mathbf{X}_n \xrightarrow{p} \mathbf{X}$, then $\mathbf{X}_n^{-1} \xrightarrow{p} \mathbf{X}^{-1}$, since inverses are generally continuous maps. You will see the power of this result shortly.

Finally, we will also make heavy use of a different, weaker notion of convergence than convergence in probability, namely [convergence in distribution](#). We say that for a sequence of random variables $X_n \xrightarrow{d} X$ whenever:

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$$

at all x at which $F_X(x)$ is continuous. Here $F_X(x)$ is the CDF of the random variable X . F_X is often called the [asymptotic distribution](#) of X_n . Convergence in distribution is sometimes written as $X_n \Rightarrow X$.

It turns out that if $X_n \xrightarrow{p} X$, then $X_n \xrightarrow{d} X$, i.e., convergence in probability implies convergence in distribution. You can check Wikipedia for a simple proof of this fact. In this sense convergence in distribution is weaker than convergence in probability. The difference at first may be a little confusing, so here is a trivial examples where convergence in distribution holds but not probability. Suppose that $X_n \xrightarrow{p} X$, where $X \sim N(0, 1)$. Then $Y = -X$ is also $\sim N(0, 1)$. You can show that $X_n \xrightarrow{d} Y$, but X_n does not converge in probability to Y . Intuitively, X and Y share the same distribution but do not correspond to the same mapping from the sample space to the real line (recall the definition of a random variable).

Slutsky and continuous mapping theorems apply to convergence in distribution as well, so we know that e.g., $X_n \xrightarrow{d} X$ implies $g(X_n) \xrightarrow{d} g(X)$ for appropriately continuous g , and if $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$, then $X_n + Y_n \xrightarrow{d} X + c$. This will also be very useful.

As a comprehensive example, let's consider a straightforward estimate of the sample variance:

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})^2$$

for iid $X_1, \dots, X_n$ from a distribution with finite moments. We can re-arrange this estimator as:

$$\begin{aligned} \hat{\sigma}^2 &= \frac{1}{N} \sum_{i=1}^N (X_i - \mu + \mu - \bar{X})^2 \\ &= \frac{1}{N} \sum_{i=1}^N (X_i - \mu)^2 + (\mu - \bar{X})^2 + 2\frac{1}{N}(\mu - \bar{X}) \sum_{i=1}^N (X_i - \mu) \end{aligned}$$

Consider the random variable $Y_i = (X_i - \mu)^2$. This has mean $Var(X_i)$ and finite variance (a fourth moment in this case) by assumption. Thus the law of large numbers implies that $\frac{1}{N} \sum_{i=1}^N (X_i - \mu)^2 \xrightarrow{p} Var(X)$. We know that $\bar{X} \xrightarrow{p} \mu$, so Slutsky and the CMT tells us that $\bar{X} - \mu \xrightarrow{p} 0$ and $(\bar{X} - \mu)^2 \xrightarrow{p} 0^2 = 0$. Finally, note that the final term involves $2(\bar{X} - \mu)$, which converges in probability to zero, and $\frac{1}{N} \sum_{i=1}^N (X_i - \mu)$, which by the LLN converges to zero as well. Thus by Slutsky the product converges to zero. $\hat{\sigma}^2$ is therefore a consistent estimator of σ^2 .

Inference and hypothesis testing

The process of [statistical inference](#) is how we make judgements about an estimand based on the properties of our estimators and their realizations. For example, suppose we have a realization of an estimator $\hat{\mu}$. What can we say about μ based on this result? How confident can we be that μ is one or zero or any other number for that matter?

The first way to do this is to consider a [hypothesis test](#). To define a hypothesis test, we need to define a [null hypothesis](#). This is the baseline state of the world we will assume to be true unless the data suggest otherwise. Null hypotheses on specific estimands are frequently denoted with a zero, such as $H_0 : \mu = 0$. The value of the estimand under the null is sometimes also written just μ_0 . To conduct a hypothesis test, we devise a [testing procedure](#). This is a rule that computes a [test statistic](#), which is a function of the sample $T(X)$, and decides when to reject. A [critical value](#) is a threshold value of the test statistic above which you decide to reject the null. A testing procedure is associated with a [size \(or level\)](#), which tells us the probability of falsely rejecting the null—i.e., making a [type one error](#). That is, size is given by $Pr(T(X) > t | H_0)$.

Suppose, for example, we are studying an iid sample from a $N(\mu, 1)$ distribution and wish to conduct inference on μ . You can show based on properties of normals discussed in the last lecture that the usual sample mean estimator $\bar{X}_N \sim N(\mu, 1/N)$. Under the null, we know therefore that $\bar{X}_N \sim N(\mu_0, 1/N)$. This is called the [sampling distribution](#) under the null. We can use this distribution to assess how likely we'd be to see certain realizations of $\bar{X}_N$ under the null. Suppose, for example, we have a rule that says reject μ_0 when $\sqrt{N}|\bar{X}_N - \mu_0| > 1.96$. Based on the sampling distribution of $\bar{X}_N$, we know that $Pr(\sqrt{N}|\bar{X}_N - \mu_0| > t) = 2\Phi(-t)$. We can therefore conclude that the size of this test is approximately 0.05. If you wanted to construct a test with different size, you could use a different critical value (e.g., 2.576 corresponds to one percent test).

We may also be interested in a testing procedure's [type two error](#) rate, which is the probability of *not* rejecting the null when it is false. The [power](#) of a testing procedure is inversely related to the type two error rate, so that high power makes type two error less likely. You can see that power depends on the [alternative hypothesis](#), sometimes denoted H_1 . We know that in our normal example a rule that rejects when $\sqrt{N}|\bar{X}_N - \mu_0| > 1.96$ will have 10% size. Suppose we are testing $H_0 : \mu = 0$. If in reality $\mu = 100$, then realizations of $\bar{X}_N$ will be very far from 0 for a reasonable N , so we will be very likely to reject the null. This test is therefore very powered against this alternative. But if $\mu = 0.0001$, we would need a substantial amount of data to reject $\mu = 0$, so the test is not very powered. The process of [power calculations](#) in experiments is exactly about this trade off: for a test with a fixed size, how much data do I need to collect to have a reasonable power against specific alternatives?

Rather than reporting the results of a specific testing procedure, it is far more common to report a [p-value](#) associated with a test. This p-value is essentially the size of a test associated with treating $T(N)$ as the relevant critical value. You can also think of this as the largest sized test that would reject $T(X)$. In our running example, the p-value would be the size associated with a test with critical value $T(N) = \sqrt{N}|\bar{X}_N - \mu_0|$. From our work above, for any realization of $\bar{X}_N$ we know this probability is just $2\Phi(-\sqrt{N}|\bar{X}_N - \mu_0|)$.

For formally, we define a p-value as a test statistic constructed from the sample $P(X)$ that satisfies $Pr(P(X) \leq \alpha | H_0) \leq \alpha \forall \alpha \in [0, 1]$, where the H_0 conditioning indicates that the probability is assessed under the null. In words, we should see a p-value smaller than α no more than α percent

of the time under the null. You can see that $P(X) = 2\Phi(-\sqrt{N}|\bar{X}_N - \mu_0|)$ is a valid p-value in our running example because it measures the mass in the tails of the sampling distribution of $\bar{X}_N$. Under the null, the likelihood of seeing a p-value at least this small is the likelihood of seeing a realization of $\bar{X}_N$ in these tails, which is exactly $P(X)$. If the null is that $\mu_0 = 0$ and our data tells us $\bar{X}_N = 0.1$, the p-value is $2(1 - \Phi(\sqrt{N}/10))$. For a small sample like $N = 10$, the p-value would be about 0.75. In words, this tells you that under the null, seeing a mean of 0.1 with a sample size of 10 is not that unusual. If we had $n = 500$, however, this would be a much more rare realization of $\bar{X}_N$ under the null. The p-value is about 0.025.

P-values are confusing, possibly for everyone. People often interpret them incorrectly as “the probability that the null hypothesis is true.” This is wrong, since that statement requires assessing something like $Pr(H_0|data)$, as opposed to $Pr(data|H_0)$, which is what a p-value does. Despite the fact that p-values are usually continuous, the profession also fixates on a number of discrete thresholds. Results get extra attention when p-values are less than 0.1, even more when they are less than 0.05, and quite a bit more when they fall below 0.01. When a p-value falls below a particular threshold, we say that we can reject the null at the 10/5/1 percent level. But it is up to you as the researcher to decide what level of “statistical significance” is important to you. That is part of creating a testing procedure.

Hypotheses tests do not have to be constructed in the two-sided way I did above. We can, for example, consider a [one-sided hypothesis test](#) such as $H_0 : \mu \leq 0$, $H_1 : \mu > 0$. Here we need to take care that the value satisfies $Pr(P(X) < \alpha|\mu) \leq \alpha \forall \mu \in H_0$, so that it still lets us control type one error under the null. This was trivial in the case where H_0 was a singleton. When the null is that $\mu \leq 0$ we want to take $\sup_{\mu \in H_0} Pr(T > t|\mu)$, which usually implies putting μ on the boundary of the parameter space defined by the null. In our normal example, this amounts in this case to using $\Phi(-\sqrt{N}t)$.

The other way to summarize what we know about an estimand from the data is by constructing [confidence intervals](#). A confidence interval is a rule that constructs a set in which we expect the estimand to fall in with pre-specified probability. That is, we are looking for a set C_n such that $Pr(\mu \in C_n) \geq 1 - \alpha$. The common way to construct C_n is through [test inversion](#). One can simply collect all μ such that our testing procedure fails to reject at the α level. In our a normal example, a 95% confidence interval can be constructed as:

$$C_n = \bar{X}_N \pm \frac{1.96}{\sqrt{N}}$$

As an [exercise](#), you should show a confidence intervals based on inverting a level- α test contains the truth with probability $\geq 1 - \alpha$.

It is also very common to summarize the uncertainty about estimands through the use of a [standard error](#). A standard error of an estimator is an estimator of the standard deviation of its sampling distribution. Sometimes this sampling distribution will be known, so that this estimator is very good. In the normal example, the standard error is simply $1/\sqrt{N}$. Other times you the standard error will not be known with certainty. It is common to report estimated parameters along with standard errors in parentheses. You can see that in the normal example, confidence intervals and standard errors are directly related, in the sense that a 95% confidence interval can be constructed as roughly $\pm$ twice the standard error.

Since the purpose of an estimator $\hat{\mu}$ is to estimate μ , we typically like estimators with small standard errors and confidence intervals. The [precision](#) of an estimator refers to how much sampling variability it has, and estimators that are more precise are (loosely) said to be more [efficient](#). We will make this more precise in a little bit.

Asymptotic approximations

You should have noticed that conducting hypothesis tests and constructing confidence intervals and standard errors required knowing the sampling distribution of the estimator. This is rarely possible in practice without making *strong* assumptions, such as that $X_i \sim N(\mu, \sigma^2)$. Luckily, under much weaker assumptions, we can use [asymptotic approximations](#) to argue that with sufficient sample sizes many estimators will be approximately normally distributed. We can then use the normal distribution to conduct hypothesis tests and do inference, so long as we're convinced this approximation is reasonably good.

Asymptotic arguments are built on [central limit theorems](#). The CLT for iid data states that for $X_1, \dots, X_n$ iid with finite variance, then:

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N (X_i - E[X_i]) \xrightarrow{d} N(0, \text{Var}(X_i))$$

It's worth stopping to admire this weird and seemingly mysterious result. We did not need to assume much about the distribution of X . In particular, it definitely need not be normal itself. Nevertheless, as we collect more data the sampling distribution of $\sqrt{N}\bar{X}_N$ will get arbitrarily close to a normal with mean zero and variance equal to the variance of X . Your first problem set will take you through an exercise that provides some intuition for this result.

When an estimator $\hat{\mu}$ satisfies $\sqrt{N}(\hat{\mu} - \mu) \xrightarrow{d} N(0, V)$, we say that $\hat{\mu}$ is “root-N” consistent. Notice that root-N consistency implies convergence in distribution of $\hat{\mu}$ to μ , but also more, since it also tells us the limiting distribution is normal.

Often our estimators are often functions of sample means. Luckily, a variation of the CLT also covers this case. The [Delta method](#) says that if $\sqrt{N}(\hat{\mu} - \mu) \xrightarrow{d} N(0, V)$, then $\sqrt{N}(g(\hat{\mu}) - g(\mu)) \xrightarrow{d} N(0, g'(\mu)^2 V)$ when g is differentiable at μ . Both the CLT and the Delta method apply to vector-valued random variables and means as well. In that case V is a matrix, and the delta method says that $\sqrt{N}(g(\hat{\mu}) - g(\mu)) \xrightarrow{d} N(0, G(\mu)VG(\mu)')$, where G is the Jacobian of g at μ (i.e., the matrix of first derivatives).

The Delta method is also a little mysterious, but its straightforward to prove using the tools we've already covered if we consider a continuous g . The mean value theorem says that:

$$g(\hat{\mu}) = g(\mu) + g'(\bar{\mu})(\hat{\mu} - \mu)$$

for some $\bar{\mu}$ between $\hat{\mu}$ and μ . Multiplying through by $\sqrt{N}$ and re-arranging shows that:

$$\sqrt{N}(g(\hat{\mu}) - g(\mu)) = g'(\bar{\mu})\sqrt{N}(\hat{\mu} - \mu)$$

We know that $\sqrt{N}(\hat{\mu} - \mu) \xrightarrow{d} X$, where $X \sim N(0, V)$. A version of the squeeze theorem applies in probability. In particular, since we know that $\hat{\mu} \xrightarrow{p} \mu$ and $|\bar{\mu} - \mu| \leq |\hat{\mu} - \mu|$, $\bar{\mu} \xrightarrow{p} \mu$ as well. Slutsky

thus tells us that $g'(\bar{\mu})\sqrt{N}(\hat{\mu} - \mu) \xrightarrow{d} g'(\mu)X$. Since $g'(\mu)$ is a constant, this has distribution $N(0, g'(\mu)^2 V)$.

To conclude, let's put everything together by revisiting the case of an iid sample $X_1, \dots, X_N$ from distribution F with unknown mean μ and variance σ^2 . We want to test the null hypothesis $\mu_0 = 0$, but are unwilling to assume that F is normal or that we know its variance.

Consider:

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N X_i, \quad \hat{\sigma} = \sqrt{\frac{1}{N} \sum_{i=1}^N (X_i - \hat{\mu})^2}$$

$$\hat{\theta} = \hat{\mu} / \hat{\sigma}$$

We know that under the null $E[\hat{\mu}] = \mu = 0$. Thus $\sqrt{N}\hat{\mu} \xrightarrow{d} N(0, \sigma^2)$ by the central limit theorem. We also showed above that $\hat{\sigma}^2 \xrightarrow{p} \sigma^2$. Continuous mapping tells us that the square root of $\hat{\sigma}^2$ converges to σ . $\sqrt{N}\hat{\theta}$ thus converges by Slutsky to a $N(0, 1)$ distribution.

This is useful because we've pinned down the asymptotic distribution of $\sqrt{N}\hat{\theta}$. When N is large, we expect it to look a lot like a $N(0, 1)$ distribution. Note that this means that $\hat{\theta}\hat{\sigma} = \hat{\mu}$ will be distributed as $N(0, \hat{\sigma}^2/N)$. We can therefore conduct inference pretending the sampling distribution of $\hat{\mu}$ is exactly this distribution. You should run some simulations to convince yourself that this is a reasonable thing to do for a variety of distributions F .

When an asymptotic approximation kicks in is hard to know because it depends on the underlying distributions. If F is already normal, things are going to look pretty normal with a relatively small N . For other distributions, you may need several hundred observations before the approximation becomes "good." If we have time later in this course, will discuss this and other ways to quantify our uncertainty that don't rely on these asymptotic approximations.

Econometricians generally distinguish what they can say about the "finite" sample behavior of an estimator relative to its "asymptotic" behavior. It is very possible for an estimator to have a nice asymptotic distribution with a reasonable variance but that behaves poorly when one is equipped with a finite amount of data. I will try to illustrate this idea further as we go.